MIL-D-51188C(EA)
29 August 1977
SUPERSEDING
MIL-D-51188B(MU)
24 September 1968

MILITARY SPECIFICATION

DISPERSER, RIOT CONTROL AGENT,

HELICOPTER OR VEHICLE-MOUNTED, M5

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of disperser.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-781	-	Steel Strapping, Flat.
TT-W-572	-	Wood Preservative.
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-T-60	-	Tape; Pressure-Sensitive Adhesive, Waterproof, for Packaging.

FSC 1040

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Comment, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-P-130 - Paper, Wrapping, Laminated and Creped.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-C-17504 - Coating Compound, Acrylic Clear.
MIL-I-22110 - Inhibitors, Corrosion, Volatile, Crystalline.

STANDARDS

FEDERAL

FED-STD-356 - Commercial Packaging of Supplies and Equipment.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

EDGEWOOD ARSENAL

DL116-6-166 - Disperser, Riot Control Agent, Helicopter or Vehicle-Mounted, M5.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and requirements.

3.1.1 Materials. All materials cited on Drawing D116-6-166 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

3.1.2 Components. All components of the disperser shall conform to the specifications and drawings listed on Drawing D116-6-166 and on the subsidiary drawings.

3.2 Assembly. The disperser shall be assembled as shown on Drawing E116-6-166.

3.3 Functioning and leakage (see 6.3).

3.3.1 Air. The disperser shall function and not leak when the high pressure system is charged with 2000 ± 100 pounds per square inch gage (psig) (907 ± 7.2 kilograms per square centimeter) (kg/cm^2) of air and tested in accordance with 4.4.4.

3.4 Serial number. Each disperser shall be identified with a serial number as specified in the contract or order (see 6.2).

3.5 Preproduction. Prior to the start of regular production, a preproduction sample of dispersers shall be produced in accordance with this specification for examination and tests (see 4.3).

3.6 Workmanship. The disperser shall be free from contamination (oil, grease or dirt) and damage such as cracks, dents or deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of two dispersers shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination and test. The sample dispersers shall be examined and tested in accordance with the classification of defects (4.4.3.3). The AQL's do not apply.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The acceptance number is zero. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the dispersers produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and test. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.3 Inspection procedure.

4.4.3.1 For examination and test. The sample dispersers shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.

4.4.3.2 For critical defects. Each item in the lot shall be tested for its critical characteristic listed in the classification of defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

4.4.3.3 Classification of defects.

(a) Disperser, riot control agent, helicopter or vehicle-mounted, M5 (Drawing E116-6-166).

Categories		Defects	Acceptance standards
Critical:			
1		Functioning and leakage	4.4.4
Major:	AQL	1.0 percent defective	
101		Component missing or incorrectly assembled	
102		Serial number missing	
103		Workmanship (3.6)	

(b) Support assembly (Drawing E116-6-178).

Categories		Defects
Critical:		None defined
Major:		1.0 percent defective
101		Component missing or incorrectly assembled
102		Diameter and location of guides incorrect
103		Welding incorrect
104		Perpendicularity incorrect
105		Diameter (.207) holes and location incorrect
106		Gage holes missing
107		Workmanship (3.6)
Minor:	AQL	4.0 percent defective
201		Protective finish missing or incorrect

(c) Base assembly (Drawing D116-6-181).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Component missing or incorrectly assembled
102		Diameter and location of container holes incorrect
103		Diameter and location of support holes incorrect
104		Parallelism of legs to base incorrect
105		Welding incorrect
106		Workmanship (3.6)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

4.4.4 Functioning and leakage test. The test shall be conducted as follows: Pressurize the high pressure system with the required pressure using clean dry air. Open the high pressure valve. The low pressure gage shall indicate 45 to 55 psig (3.285 to 3.866 kg/cm^2). Slowly open the low pressure valve. When the low pressure valve is fully open, the low pressure gage shall still indicate the above range. Using a soap solution or suitable leak detecting solution, examine all joints and fittings for leakage. Fully open and close the ball valve three times to allow short bursts (open valve for approximately 5 seconds) of air to escape. After each burst of air, leave the valve closed long enough for the low pressure to stabilize. The low pressure gage shall indicate the above range. Open the ball valve to expel all the air. When air no longer escapes from the low pressure system, open the charging valve to expel any excess air from the high pressure system.

5. PREPARATION FOR DELIVERY

5.1 Cleaning and drying. The disperser and components shall be cleaned and dried in accordance with the applicable requirements of MIL-P-116. Cleaning method C-1 and an appropriate drying method listed in cited specification shall be used.

5.2 Preservation and packaging.

5.2.1 Level A.

(a) Repainting. Damaged portions of painted areas shall be treated and repainted in accordance with MIL-T-704, type A, the color of the finish coat shall conform to color olive drab, number 24087 or 34087 of Fed-STD-595.

(b) Preservation and packaging methods. Unless otherwise specified, preservatives including application and packaging methods cited herein shall conform to MIL-P-116.

5.2.1.1 Unit preservation and packaging.

(a) Disperser assembly. The assembled system with all valves open shall be purged with dry nitrogen. With the ball valve closed, the assembled system shall be charged to 5 psig (0.3515 kg/cm^2 with dry nitrogen after which the high and low pressure shut-off valves and the charging valve shall be closed. A warning tag conforming to type B, Class 1 or 2 of UU-T-81 shall be affixed in a conspicuous place on the unit, stating that the system has been charged with the above pressure with dry nitrogen and instructing to discharge the nitrogen before using by opening

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

the high pressure, pressure relief and ball valves. All openings into the unit shall be sealed with type III, Class 1 pressure sensitive tape conforming to PPP-T-60. The gage glass shall be protected with fitted fiberboard discs from type CF or SF class-weather resistant fiberboard conforming to PPP-F-320, secured in place with the above cited tape. All exposed unpainted surfaces including threads shall be coated with type P-7 preservative oil. The excess oil shall be wiped off. All metal name plates and both sides of warning tag shall be coated with acrylic coating compound conforming to MIL-C-17504.

(b) Hose assembly. The hose shall be wrapped with Grade A, greaseproof, barrier material conforming to MIL-B-121 or wrapping paper conforming to MIL-P-130 and the wrap secured in place by either tying or taping. The two lengths of hose shall then be individually coiled. Each coil shall be secured in at least three places by either tying or taping.

(c) Gloves, rubber. Rubber gloves shall be sprinkled with powdered talc and then packaged in accordance with method IC-1 or IC-3.

(d) Hood, gas mask. The hood shall be laid face up with bottom edges aligned and straps folded over the hood. Two hoods alternately reversed shall be placed in a paperboard folding box conforming to PPP-B-566 and packaged in accordance with method IC-2.

(e) Straps, webbing. Straps shall be packaged in accordance with method III.

(f) Adapters, bushing, coupling, hose tie downs, helicopter tie downs, snaphook and ring assembly. These items shall have threads and other exposed unpainted areas coated with type P-7 preservative oil. The exposed threads of the adapters and bushing shall be protected by fitted plastic sleeves or covered with pressure-sensitive tape. These items together with the packaged rubber preformed packing (g) and the packaged rupture discs (h) shall be packaged in accordance with Method IC-2.

(g) Packing, preformed, rubber. The packing shall be packaged in accordance with method IC-1 or IC-3. To protect the packing, two stiffeners, fabricated from type CF or SF fiberboard conforming to PPP-F-320 and of the same size as the inside bag dimensions shall be placed in the bag with the packing between the two stiffeners.

(h) Discs, rupture. The discs shall be packaged in either a plastic or metal box. Cushioning shall be used as required to prevent movement of the discs in the box.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

(i) Technical publication. Publication shall be packaged using method IC-1 or IC-3.

(j) Accessory agent tanks. Fifteen grams of type I, crystalline volatile corrosion inhibitor conforming to MIL-I-22110 shall be applied on the internal surface of the tank. To provide adequate coverage on such surfaces, the crystals shall be applied by dusting. The shipping cover plate shall be secured over the opening with clamp provided.

5.2.1.2 Intermediate packaging. Items unit packaged as specified in (b), (c), (d), (e), (f) and (i) shall be intermediately packaged together in a fiberboard box conforming to V3c of PPP-B-636. The box shall fit the contents snugly, utilizing appropriate weather-resistant filler material as necessary to restrict free movement of the unit packages. The box shall be closed by gluing in accordance with closures for weather-resistant boxes in the appendix of PPP-B-636. All seams and joints shall be sealed with tape in accordance with the water-proofing requirements of PPP-B-636.

5.2.2 Commercial packaging. The disperser and components shall be packaged in accordance with FED-STD-356.

5.3 Packing.

5.3.1 Level A.

5.3.1.1 Disperser assembly. The disperser and components including the accessory agent tank and other accessories and spare parts, preserved and packaged as specified in 5.2, shall be packed in a box conforming to Class 2, style 7, type 3 load of PPP-B-621. The disperser assembly shall be firmly secured to the skid base. The spare parts and accessories (5.2.1.2) shall be firmly blocked and braced in the box to prevent free movement. All wood components of the box shall be preservative treated by immersing in type II, composition A of TT-W-572 for a minimum period of 3 minutes. The box shall be closed and strapped in accordance with the appendix to PPP-B-621. Strapping shall conform to Class B of QQ-S-781. Inside dimensions of the box shall be approximately 37 by 24-1/4 by 46-1/2 inches in length, width and depth respectively.

5.3.1.2 Pilot pack. Unless otherwise specified (see 6.2) a pilot pack shall be prepared in accordance with 5.3.1.1 and tested in accordance with the applicable rough handling tests as specified in MIL-P-116. The tested pilot pack shall then be submitted to determine compliance with the packing requirements of this specification.

5.3.2 Level B. The disperser and components, packaged as specified in 5.2, shall be packed as specified in 5.3.1 except that domestic type or class containers and coated finish strapping shall be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

5.3.3 Commercial The disperser and components shall be packed in accordance with FED-STD-356.

5.4 Marking. In conjunction with appropriate information provided in the contract or order (see 6.2) marking shall be as specified herein. Information requirements shall be interpreted in accordance with MIL-STD-129.

5.4.1 Marking levels A and B. In addition to any special markings required by the contract or order, marking shall be in accordance with MIL-STD-129.

5.4.2 Marking, commercial. Marking shall be in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. This item is intended for use in supressing riots and similar disturbances.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of packing and packaging required.
(c) Type of filling required.
(d) Pilot pack if required.
(e) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with production.

(f) Applicable pertinent identification marking.
(g) Applicable address marking pertaining to shipping obligation of contractor (overseas or domestic) to a POE or other domestic receiver.
(h) Shipment priority number if applicable.
(i) Transportation control number, project code number and required delivery date.
(j) When crossed retorts insignia is required.
(k) Serial numbers assigned for end item and M9 gun.

6.3 Caution. Suppliers are hereby notified that every precaution should be taken to protect personnel from injury when testing dispersers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

6.4 Component specifications. The following specifications reflect the components required for the complete assembly of the M5 Disperser.

MIL-G-51025 - Gun, Portable Riot Control Agent Disperser, M9.
MIL-H-51337 - Hose Assembly.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A143

☆U.S. GOVERNMENT PRINTING OFFICE: 1977—703-020/6192

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:52Z
Check the source to verify that this is the current version before use.